

Day 10: IPv4 Header

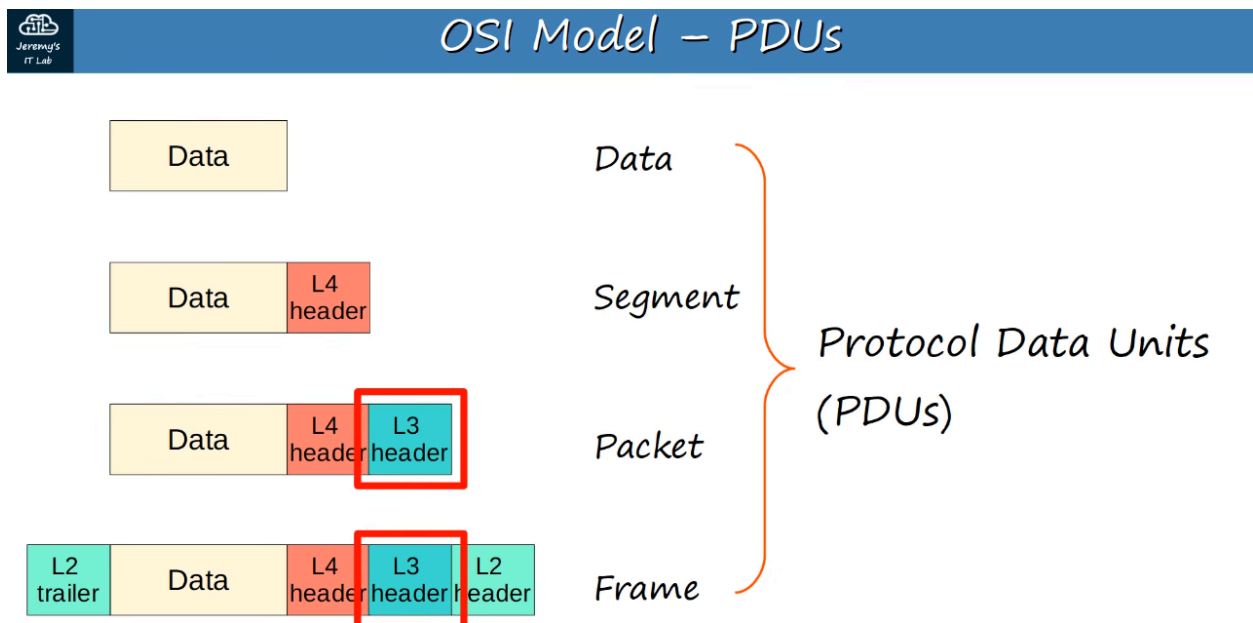
INTERNET PROTOCOL version 4 HEADER or IPv4 HEADER

HEADER is used at LAYER 3 to help send data between devices on separate networks, even on other sides of the world over the Internet.

This is known as ROUTING.

THE IPv4 HEADER is used to ENCAPSULATE a TCP or UDP Segment.

To Review:



FIELDS OF THE IPv4 HEADER

Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Version				IHL				DSCP						ECN		Total Length															
4	32	Identification																Flags		Fragment Offset													
8	64	Time To Live								Protocol								Header Checksum															
12	96	Source IP Address																															
16	128	Destination IP Address																															
20	160	Options (if IHL > 5)																															
24	192																																
28	224																																
32	256																																

FIELD	# OF BITS
VERSION	4
IHL	4
DSCP	6
ECN	2
TOTAL LENGTH	16
IDENTIFICATION	16
FLAGS	3
FRAGMENT OFFSET	13
TIME TO LIVE	8
PROTOCOL	8
HEADER CHECKSUM	16
SOURCE ADDRESS	32
DESTINATION ADDRESS	32
OPTIONS	320 Max

VERSION:

- LENGTH is 4 bits.
- Identifies the version of IP used (IPv4 or IPv6)
 - IPv4 = 0100 in Binary (Decimal 4)
 - IPv6 = 0110 in Binary (Decimal 6)

INTERNET HEADER LENGTH (IHL):

- LENGTH is 4 bits.
- The final field of IPv4 Header (Options) is variable in length so this field is necessary to indicate the total length of the header.
- Identifies the length of the header in 4-BYTE INCREMENTS.
- The MINIMUM value is 5 ($5 * 4\text{-bytes} = 20\text{ bytes}$) - Empty OPTIONS Field
- The MAXIMUM value is 15 ($15 * 4\text{-bytes} = 60\text{ bytes}$)

MINIMUM IPv4 HEADER LENGTH = 20 Bytes! MAXIMUM IPv4 HEADER LENGTH = 60 Bytes!

DSCP (Differentiated Services Code Point):

- LENGTH is 6 bits.
 - Used for QoS (Quality of Service)
 - Used to prioritize delay-sensitive data (streaming voice, video, etc.)
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ECN (Explicit Congestion Notification):

- LENGTH is 2 bits.
 - Provides end-to-end (between two endpoints) notification of network congestion WITHOUT dropping packets.
 - Optional feature that requires both endpoints, as well as the underlying network infrastructure to support it.
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TOTAL LENGTH:

- LENGTH is 16 bits.
 - Indicates the TOTAL length of the packet (L3 Header + L4 Segment)
 - Measured in bytes (not 4-byte increments like IHL)
 - Minimum value of 20 Bytes (IPv4 Header with NO encapsulated data)
 - Maximum value of 65,535 (MAXIMUM 16-bit value) = 2^{16}
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IDENTIFICATION:

- LENGTH is 16 bits.
 - If a packet is fragmented due to being too large, this field is used to identify which packet the fragment belongs to.
 - All fragments of the same packet will have their own IPv4 header with the same value in this field.
 - Packets are fragmented, if larger than the MTU (Maximum Transmission Unit)
 - The MTU is usually 1500 bytes (Max size of an Ethernet frame)
 - Fragments are reassembled by the receiving host.
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FLAGS:

- LENGTH is 3 bits
 - Used to control/identify fragments.
 - Bit 0: Reserved, always set to 0.
 - Bit 1: Don't Fragment (DF bit), used to indicate a packet that should not be fragmented.
 - Bit 2: More Fragments (MF bit), set to 1 if there are more fragments in the packet, set to 0 for the last fragment or NO fragments.
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FRAGMENT OFFSET:

- LENGTH is 13 bits
 - Used to indicate the position of the fragment within the original, unfragmented IP Packet.
 - Allows fragmented packets to be reassembled even if the fragments arrive out of order.
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TIME TO LIVE (TTL):

- LENGTH is 8 bits
- A router will drop a packet with a TTL of 0
- Used to prevent infinite loops

- Originally designed to indicated a packets maximum lifetime in seconds.
 - In practice, indicates a 'hop count': each time the packet arrives at a router, the router decreases the TTL by 1.
 - Recommended default TTL is 64.
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PROTOCOL:

- LENGTH is 8 bits
- Indicates the protocol of the encapsulated Layer 4 PDU
- Value of 1 : ICMP
- Value of 6 : TCP
- Value of 17 : UDP
- Value of 89 : OSPF (Dynamic Routing Protocol)
- List of protocol numbers on Wikipedia : [List of IP Protocol Numbers](#)

HEADER CHECKSUM:

- LENGTH is 16 bits
 - A calculated checksum used to check for errors in the IPv4 header.
 - When a router receives a packet, it calculates the checksum of the header and compares it to the one in this field of a header.
 - If they do not match, the router drops the packet.
 - Used to check for ERRORS only in the IPv4 Header.
 - IP relies on the encapsulated protocol to detect errors in the encapsulated data.
 - Both TCP and UDP have their own checksum fields to detect errors in the encapsulated data.
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SOURCE and DESTINATION:

- LENGTH is 32 bits each
- SOURCE IP = IPv4 ADDRESS of the Sender of the Packet.

- DESTINATION IP = IPv4 ADDRESS of the intended Receiver of the Packet.
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OPTIONS:

- LENGTH is 0-320 bits
- Optional / Rarely Used
- If the IHL field is greater than 5, it means that Options are present.